



東北大學

自強不息·知行合NEU

Frontier Technologies and Applications of Traversing

导线测量的前沿技术及其应用领域

Reporter: ZhangYang

Time: 2025/11/4

? Question:

Why do we need new technologies?

» low efficiency

Relying on manual aiming and recording is both **time-consuming and laborious**.

» environmental constraints

It is difficult to operate **in complex terrains** such as tunnels and jungles.

» data isolation

The measurement results are mostly in **independent coordinates**, making it difficult to integrate them directly into digital workflows.

➤ Core viewpoint

Faced with the extremely **high demands of** modern engineering for efficiency, precision, and digitization, traverse survey technology must integrate with new technologies to achieve self-innovation.

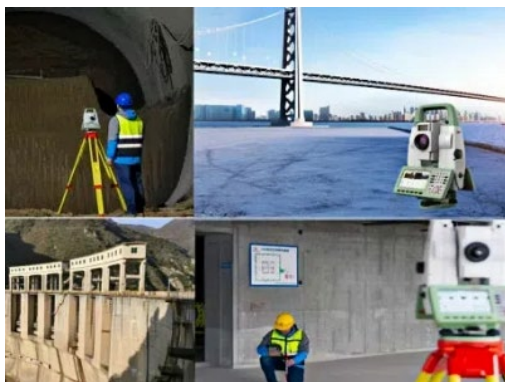


02

High-intelligence total station The king of precise surface measurement

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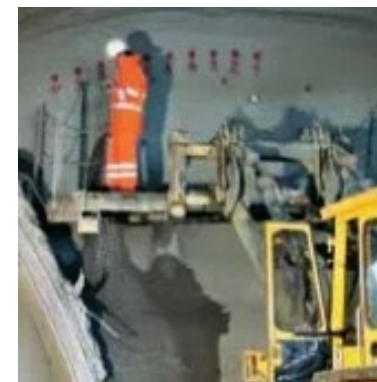
High-intelligence total station（高智能全站仪）——Leica TS16



Diverse solutions



**Rich storage
interaction functions**



**More professional
shield machine models**



**More powerful dynamic
locking performance**



**ATR Plus
automatic sighting**



**automatic height
measurement**

02

High-intelligence total station The king of precise surface measurement

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Application field of High-intelligence total station



Smart Building



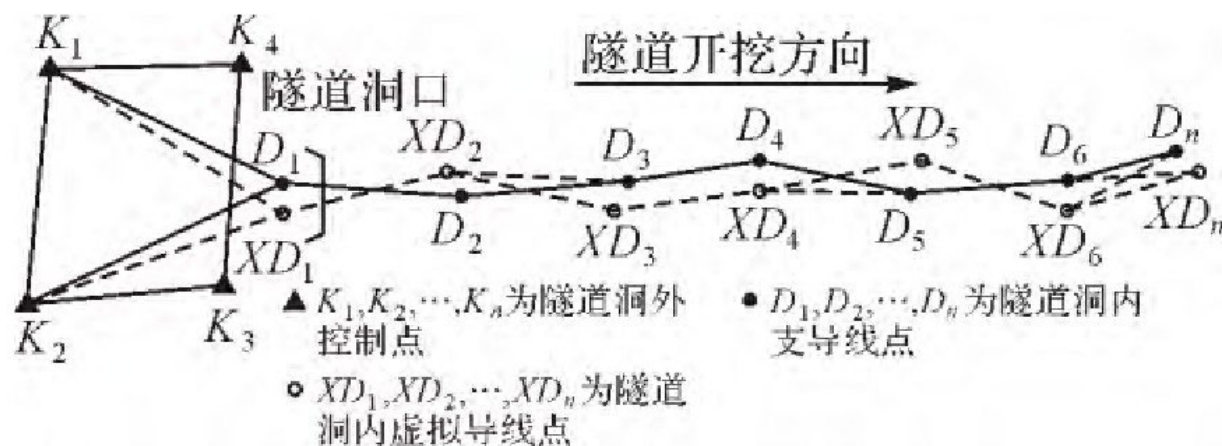
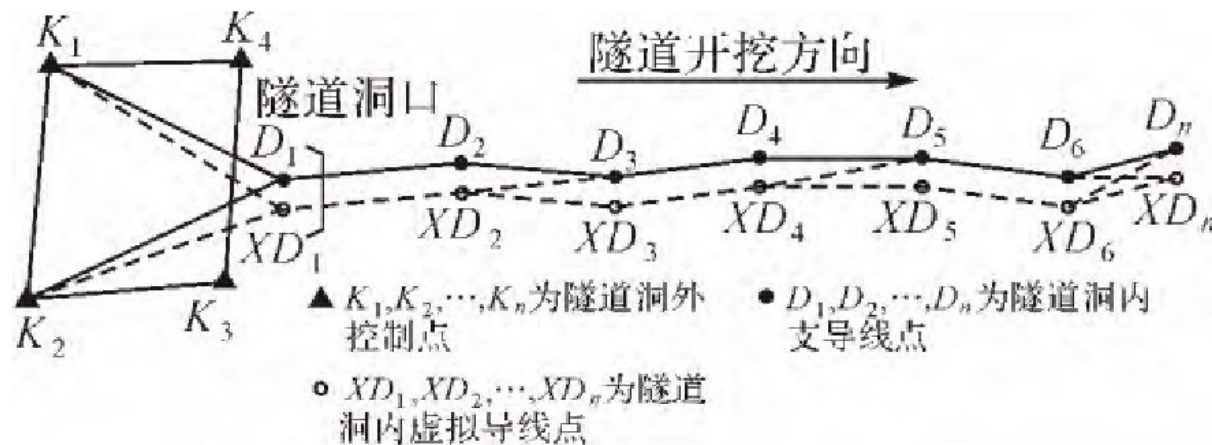
Engineering Survey



Urban Surveying

Virtual Double Traverse (虚拟双导线技术)

One real traverse + one virtual traverse generated by software via specific algorithms



➤ Economic and Efficient □

Saves nearly **half of** the pile - embedding work and overall workload.

➤ Precision Guarantee □

Achieves **check effect and precision** comparable to physical double traverses.

➤ Flexible Layout □

Particularly suitable for **narrow and inconvenient environments**.

Application field of Virtual Double Traverse



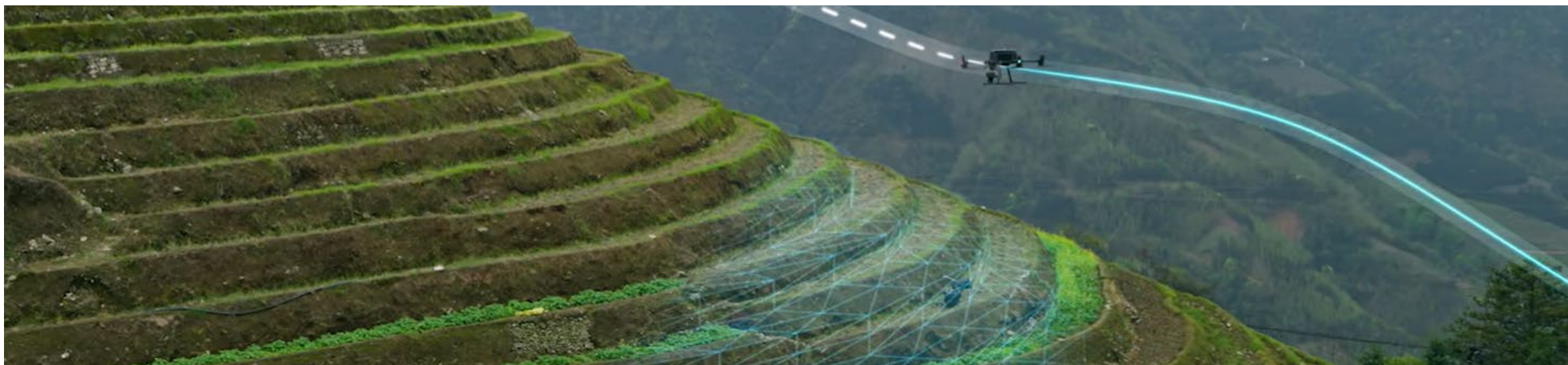
Control survey inside
long tunnels



Breakthrough survey
of subway tunnels

UAV Integrated with RTK (融合 RTK 的无人机摄影测量) —— DJI M350 RTK

Route Planning → Automatic → Image Acquisition → Real-Scene 3D Model Generation



Extremely High Efficiency

Completes square-kilometer-level data acquisition **in a short period of time**.

Safe and Non-Contact

Keeps away from **hazardous areas** (high-voltage lines, slopes, mining areas).

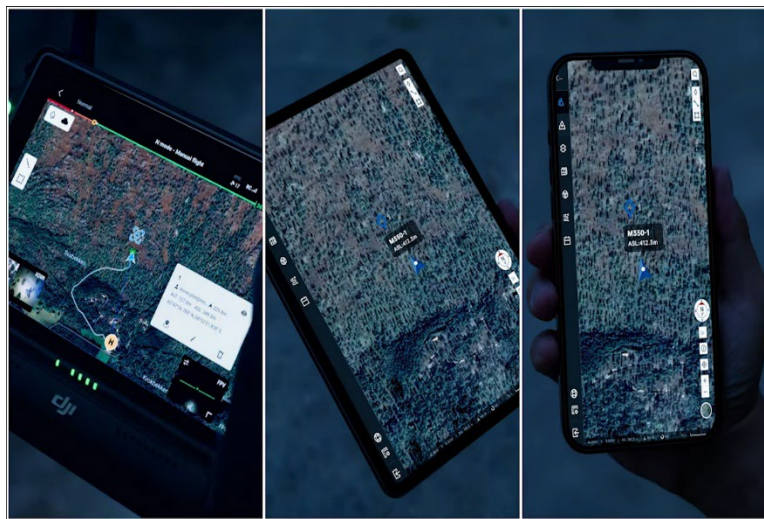
Rich Data Results

TDOM (真正射影像), **DSM** (数字表面模型), **Real-Scene 3D Model** (实景三维模型), **Point Cloud** (点云).

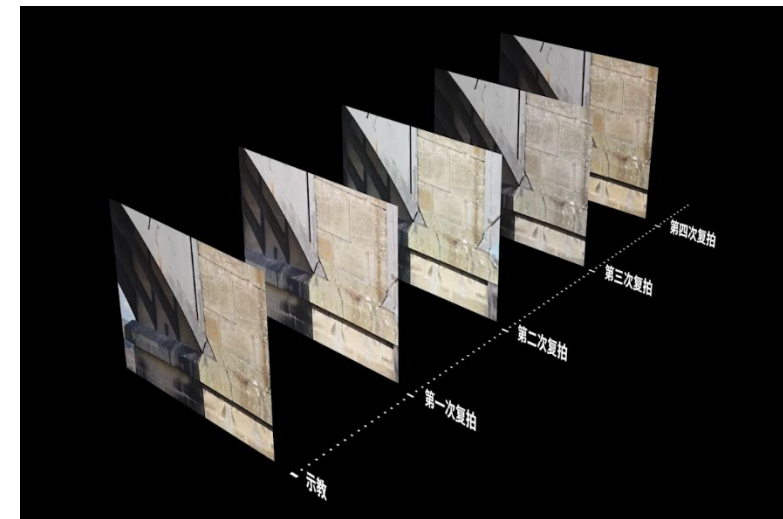
Application field of UAV Integrated with RTK



High-Precision Surveying
and Mapping



Air-Ground
Collaborative Operation

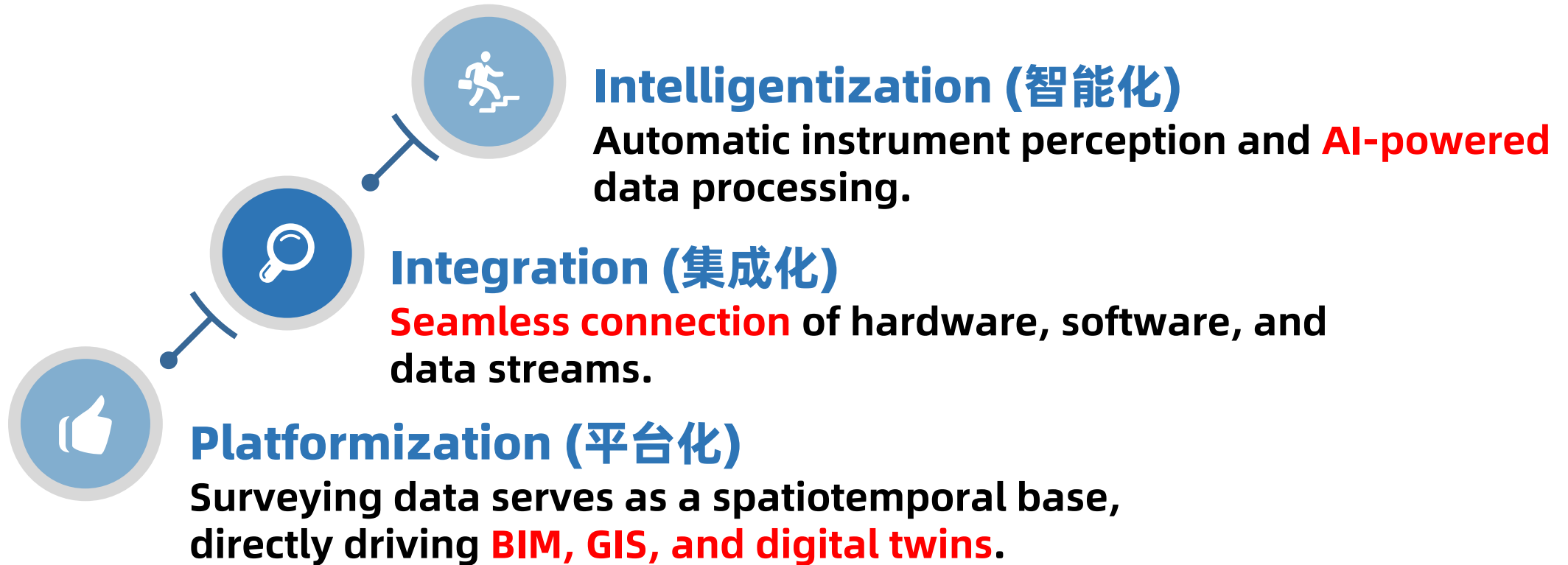


Automated Fine
Inspection

Core Features Comparison of the Three Technologies

Technology	Core Advantages	Optimal Application Scenarios
High-Intelligence Total Station	Highest absolute accuracy, automation	Precision engineering setting-out , automated deformation monitoring
Virtual Double Traverse	Economical , reliable, powerful algorithms	High-grade control networks in underground/restricted spaces
UAV RTK	Extremely high efficiency , wide coverage, safety	Large-scale topographic surveying and mapping, 3D modeling , inspection

Marching Towards Surveying and Mapping Future



By and Large: Mastering these cutting-edge technologies means mastering the core competitiveness of future engineering surveying.

References

- [1]沙红良,陆德中,付康,等. 融合无人机LiDAR与摄影测量的河道地形测绘方法[J].人民长江, 2023,54(04):157-162.DOI:10.16232/j.cnki.1001-4179.2023.04.022.
- [2]黄先锋,张帆,高云龙,等. 深度学习与无人机航空摄影测量融合技术[J].中国减灾,2020,(11):29.
- [3]柏文锋. 基于智能全站仪的地铁隧道自动化监测精度分析及验证[J].测绘通报,2018,(11):107-110.DOI:10.13474/j.cnki.11-2246.2018.0361.
- [4]郭平,王靠省,李小龙. 隧道洞内虚拟双导线测量技术研究[J].测绘通报,2016,(11):89-92. DOI:10.13474/j.cnki.11-2246.2016.0373.
- [5]毛建平,周小冬,覃乐勤. 智能全站仪在大跨径桥梁荷载试验中的应用[J].西部交通科技,2016,(09):87-89+105.DOI:10.13282/j.cnki.wccst.2016.09.023.
- [6]潘华志.智能全站仪动态测量与数据处理方法研究[D].解放军信息工程大学,2007.
- [7]卫建东.智能全站仪变形监测系统及其在地铁结构变形监测中的应用[D].中国人民解放军信息工程大学,2002.



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Thank you all for watching my presentation

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